

HEALTHCONNECT CLINIC EXPERIENCE LAB

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

WEEK 8: Final Integration → Project Readiness → Presentation → Showcase

PROJECT CONTINUITY

My Week 8 work continues directly from Weeks 4–7 using the same HealthConnect Appointment Dataset. No new project or dataset was introduced.

- **Week 4:** Problem Understanding, Resource Review and Solution Planning
- **Week 5:** Analysis, Dashboard Development and Initial Implementation
- **Week 6:** Integration Advanced Development Validation
- **Week 7:** Testing, Refinement and End-to-End Validation
- **Week 8:** Final Integration Project Readiness Presentation Showcase

The final contribution is the validated Power BI Data Analytics Dashboard and Final Analytics & Decision Support Package.

WEEK 7 → WEEK 8 TRANSITION

Main Week 7 Output

I completed and tested the HealthConnect Power BI Data Analytics Dashboard with validated KPIs, visualizations, filters and high-risk segmentation.

Most Important Testing Result

The dashboard results matched the 5,000 records, with 2,314 Attended + 2,423 No-Shows + 263 Cancelled = 5,000.

Major Issue Discovered

The **61.02%** no-show rate for the broader 2+ previous no-show group was initially confused with the combined high-risk rate.

Refinement

The high-risk definition was corrected to Previous No-Shows ≥ 2 AND Booking Lead Days ≥ 31 . This produced 233 appointments, 170 no-shows and a 72.96% no-show rate.

Component Ready

The Power BI dashboard and final analytics package are validated and ready for integration and presentation.

Remaining Integration

The validated analytics findings need to be connected with the Data Science, ML Engineering, GenAI and Project Management components.

Final Presentation

I will demonstrate the problem, dataset, KPIs, key no-show patterns, high-risk segment, dashboard, recommendations, limitations and cross-track contribution.

FINAL INTEGRATION READINESS

Final Component

My final component is the HealthConnect Power BI Data Analytics Dashboard and Final Analytics & Decision Support Package.

Problem Addressed

It addresses missed appointments by identifying observed no-show patterns and appointment groups requiring further monitoring and patient-support consideration.

Current Status

The component is validated and ready for final integration and presentation.

Week 7 Improvements

I validated KPIs, dashboard figures, slicers and visualizations corrected the high-risk calculation, reviewed data quality and identified appropriate pre-appointment variables for future predictive analysis.

Dependencies

The main dependency is the final integration with Data Science, ML Engineering, GenAI and Project Management.

Tracks Depending on My Output

Data Science can use the validated analytical findings and pre-appointment variables. ML Engineering can use the analytical requirements for the technical workflow. GenAI and Project Management can use the findings and business context for the wider solution and presentation.

Output I Provide

I provide the final dashboard, KPIs, validated findings, visualisations, high-risk segmentation, business insights, recommendations, limitations and presentation materials.

Output Needed From Other Tracks

I need the validated outputs from the other tracks for final HC-POD integration. Where an output has not yet been completed, it remains an integration dependency and is not presented as completed.

Evidence of Readiness

Readiness is demonstrated by validated KPI calculations, correct totals, outcome reconciliation, tested filters, validated visualisations, corrected high-risk segmentation and documented limitations.

Remaining Limitations

The data is fictional and anonymised. Missing values remain in some variables. Sunday appointments conflict with the Knowledge Base operating schedule. Reminder-channel blanks require interpretation. The analysis shows associations, not causation, and future predictive modelling must avoid data leakage.

FINAL DATASET AND KPIs

The same HealthConnect dataset from Weeks 4–7 was used.

It contains:

5,000 appointments, 18 variables, 1 January 2025–30 June 2026, 5,000 unique appointment IDs

0 duplicate IDs

Final KPIs

Total Appointments: 5,000, **No-Shows:** 2,423, **No-Show Rate:** 48.46%, **Attended:** 2,314, **Attendance Rate:** 46.28%, **Cancelled:** 263, **Cancellation Rate:** 5.26%, **Reminder Coverage:**

72.68%, **31+ Lead No-Show Rate: 60.49%, 2+ Previous No-Show Rate: 61.02%, Combined High-Risk No-Show Rate: 72.96%**

The high-risk segment contains **233 appointments and 170 no-shows**.

FINAL VALIDATED FINDINGS

Previous No-Shows

Observed no-show rates increased from: **43.51% → 53.49% → 59.36% → 68.82%** for 0, 1, 2 and 3+ previous no-shows respectively.

Booking Lead Time

Observed no-show rates increased from **26.51%** for 0–1 days to **60.49%** for 31+ days.

Combined Risk

Appointments with **2+ previous no-shows and 31+ booking lead days** recorded a **72.96%** observed no-show rate.

Reminders

No-show rates were **47.36% with a reminder** and **51.39% without a reminder**. This is an association and does not prove reminder effectiveness.

Reminder Channels

Observed no-show rates were: SMS: **45.8%**, Email: **48.4%**, WhatsApp: **49.8%**

Appointment Type

Observed no-show rates were: Follow-up: **51.23%**, Diagnostic Test: **49.75%**, Specialist Consult: **47.44%**, General Consult: **46.64%**

Distance

The **>20 km** group recorded **57.76%**, compared with **45.28%** for the 2–5 km group.

FINAL POWER BI DASHBOARD

The final dashboard contains:

- KPI cards.
- Appointment outcome analysis.
- Previous no-show analysis.
- Booking lead-time analysis.
- Reminder status and channel analysis.
- Appointment-type analysis.
- Distance analysis.
- Previous No-Show × Booking Lead-Time heatmap.
- Risk-factor analysis.
- Supporting evidence visuals.

Slicers include **appointment type, appointment day and reminder status**.

The dashboard is designed as a **decision-support and monitoring tool**, not an automatic patient-level decision system.

KEY BUSINESS INSIGHTS AND RECOMMENDATIONS

Key Insights

1. **Missed appointments are a major issue:** The no-show rate is **48.46%**.
2. **Previous no-shows matter:** The rate reaches **68.82%** for patients with 3+ previous no-shows.
3. **Long booking lead times matter:** The 31+ day group has a **60.49%** no-show rate.
4. **Combined risk is higher:** The 2+ previous no-show and 31+ lead group has a **72.96%** no-show rate.
5. **Distance may be relevant:** The >20 km group has a **57.76%** no-show rate.
6. **Reminder status differs:** The reminder and no-reminder groups show different observed no-show rates.

Recommendations

HealthConnect should:

- Monitor patients with previous no-shows.
- Monitor appointments booked 31+ days ahead.
- Investigate the combined high-risk segment for appropriate support.
- Continue monitoring reminder coverage and channels.
- Investigate possible distance/accessibility barriers.
- Resolve the Sunday operating-day inconsistency.
- Continue using the Power BI dashboard for monitoring and decision support.

HOW THE ANALYSIS SUPPORTS HEALTHCONNECT

The analysis provides evidence for:

- Appointment monitoring.
- Reminder planning.
- Patient-support discussions.
- Risk segmentation.
- Future predictive modelling.
- Monitoring attendance patterns over time.
- Evidence-based operational decisions.

The findings are decision-support evidence and should not be treated as automatic patient-level decisions.

HC-POD CROSS-TRACK INTEGRATION

Although I didn't get anyone to collaborate with at this final stage, I documented the required cross-track dependencies and designed my Analytics output so that it can connect with the wider HealthConnect solution.

Data Analytics → Data Science:

The validated findings identify suitable pre-appointment variables such as previous no-shows, booking lead days, reminder status/channel, appointment type and distance. These can support future predictive modelling. The combined high-risk segment recorded a 72.96% observed no-show rate.

Data Analytics → ML Engineering:

The Analytics track provides validated variables and data-leakage requirements to support a reproducible prediction workflow using information available before the appointment outcome.

Data Analytics → GenAI:

The Analytics findings provide business context for administrative support use cases, while GenAI should continue to use the approved HealthConnect Knowledge Base.

Data Analytics → Project Management:

The validated KPIs, findings, recommendations, limitations and dashboard evidence provide information for final coordination, documentation and presentation.

Therefore, even where direct collaboration was not available, the final Analytics component was structured with clear inputs, outputs and dependencies so that it can be integrated into the wider HealthConnect solution.

FINAL END-TO-END WALKTHROUGH

1. Problem

HealthConnect wants to reduce missed appointments and improve patient support.

2. Data

The solution uses the 5,000-record HealthConnect Appointment Dataset, Data Dictionary and Clinic Knowledge Base.

3. Analytics

Analytics identifies patterns involving previous no-shows, booking lead time, reminders, appointment type and distance.

4. Predictive Component

Data Science can use validated pre-appointment variables to support predictive modelling.

5. ML Engineering

ML Engineering can organise the predictive process into a reproducible technical workflow while preventing data leakage.

6. GenAI

GenAI supports administrative patient interactions using the approved HealthConnect Knowledge Base.

7. Project Management

Project Management coordinates the different workstreams, dependencies and final presentation.

8. Final Solution

The overall workflow is:

Data → Analytics → Predictive Insights → ML Workflow → GenAI Support → Patient/Administrative Action

9. Business Value

The solution can help HealthConnect identify appointment groups with higher observed no-show rates, improve monitoring, support appropriate reminders, investigate accessibility issues and provide evidence for future predictive modelling.

10. Limitations

The dataset is fictional and anonymised. Missing values remain, Sunday data requires clarification, associations do not prove causation, and predictive outputs require separate validation.

FINAL ANALYTICS OUTPUT

The final **HealthConnect Final Analytics & Decision Support Package** contains:

Final Power BI dashboard, Final validated KPIs, Validated findings, Key visualizations, High-risk segmentation, Business insights, Recommendations, Analytical limitations, Cross-track information, Executive summary.

LESSONS LEARNED

Through Weeks 4–8, I learned that analytical results must be continuously validated before being used for business decisions. Week 7 showed the importance of checking KPI definitions and segmentation calculations, especially the difference between the **61.02% broader 2+ previous no-show rate** and the **72.96% combined high-risk rate**.

I also learned that correlation or association should not be presented as causation, and that cross-track collaboration requires clear outputs, dependencies and evidence. Finally, I learned that good analytics must not only explain what happened but also translate findings into practical business decisions and support the wider multidisciplinary solution.

FINAL EXECUTIVE SUMMARY

The HealthConnect Data Analytics track progressed from problem understanding in Week 4 to analysis and dashboard development in Week 5, advanced segmentation and validation in Week 6, testing and refinement in Week 7, and final integration in Week 8.

Using the same **5,000 appointment records**, the final analysis found a **48.46% no-show rate**, **46.28% attendance rate** and **5.26% cancellation rate**.

Previous no-show history and booking lead time showed increasing observed no-show patterns. The combined segment of **2+ previous no-shows and 31+ booking lead days** recorded a **72.96% no-show rate across 233 appointments**.

The final Power BI dashboard provides validated KPIs, interactive slicers, analytical visualisations, risk segmentation and business insights.

The findings support appointment monitoring, reminder planning, patient-support discussions and future predictive modelling. The solution remains a decision-support framework because the data is fictional and anonymised, some values are missing, the Sunday inconsistency remains unresolved, associations do not prove causation, and predictive outputs require further validation.

FINAL CONCLUSION

The Week 8 Data Analytics component is **validated and ready for final integration and presentation**.

It provides HealthConnect with a final dashboard, validated KPIs, no-show analysis, high-risk segmentation, business insights, recommendations, limitations and cross-track contributions.

The main analytical evidence shows that previous no-show history and longer booking lead time are associated with higher observed no-show rates. The combined high-risk segment recorded a **72.96% observed no-show rate**.